

DEVI AHILYA UNIVERSITY
INSTITUTE OF ENGINEERING AND TECHNOLOGY, INDORE

CLASS TEST -2

SUBJECT CODE - 3VLRC3

SUBJECT NAME- SURVEYING

DATE - 05/10/2023

TIME - 80 MIN.

Attempt any four of the following:

(5 marks each)

1. Explain Ranging and its types.
2. What do you mean by Bearing, Explain Various types of bearings.
3. Define- True Meridian, Magnetic meridian, Arbitrary meridian, grid meridian, magnetic declination with sketch.
4. The Magnetic Bearing of a line AB is $S 45^{\circ}15' W$. Find its True bearing, if the declination is $12^{\circ}25' E$.
5. The Fore Bearings of the lines AB, BC, CD and DE are $45^{\circ}30'$, $120^{\circ}15'$, $200^{\circ}30'$, and $280^{\circ}45'$, respectively. Find angles B, C and D.
6. Convert the following
WCB to QB. (i) $22^{\circ}30'$ (ii) $170^{\circ}12'$ (iii) $211^{\circ}54'$ (iv) $327^{\circ}24'$.
QB to WCB (i) $N 12^{\circ}24' E$ (ii) $S 31^{\circ}36' E$ (iii) $S 68^{\circ}6' W$ (iv) $N 5^{\circ}42' W$

BEST OF LUCK

Q.2. Find the position and magnitude of maximum deflection for the beam as shown in figure-2. Take $E = 200 \text{ GPa}$ and $I = 7500 \text{ cm}^4$.

2 kN/m

2 kN